Chapter 5. The Riemann–Stieltjes Integral

Riemann integral, integrability criteria, fundamental theorem, bounded variation, Stieltjes integrals

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5.1 The Riemann integral

Motivation. Area under a graph is approximated by rectangles. The Riemann integral makes this rigorous by squeezing the true area between upper and lower sums; the work is in deciding which functions allow the squeeze.

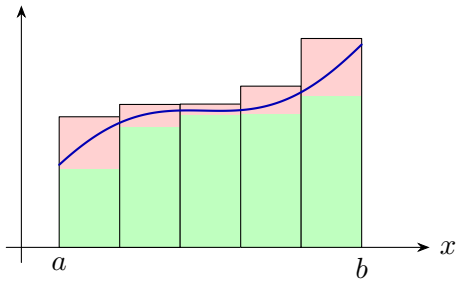


Figure. *Lower sum (green) and upper sum (green plus red) for a partition of $[a,b]$. f is integrable when refining the partition can make the red excess $U(f,P)-L(f,P)$ as small as we like.*

A **partition** of $[a,b]$ is a finite set $P=\{a=x_0<x_1<\cdots<x_n=b\}$; write $\Delta x_i=x_i-x_{i-1}$ and $\|P\|=\max \Delta x_i$ (the mesh). $Q\supset P$ is a **refinement** of P. For bounded $f:[a,b]\to\mathbb{R}$ let $M_i=\sup_{[x_{i-1},x_i]} f$, $m_i=\inf_{[x_{i-1},x_i]} f$, and

$$U(f,P)=\sum M_i\Delta x_i, \qquad L(f,P)=\sum m_i\Delta x_i.$$

The **upper** and **lower integrals** are $\overline{\int} f=\inf_P U(f,P)$ and $\underline{\int} f=\sup_P L(f,P)$; f is **Riemann integrable** ($f\in\mathcal{R}$) if they are equal, and the common value is $\int_a^b f$.

Proposition 5.1 (5.1.1)

If $P\subset Q$, then $L(f,P)\le L(f,Q)\le U(f,Q)\le U(f,P)$. For any partitions P,P', $L(f,P)\le U(f,P')$; hence $\underline{\int} f\le\overline{\int} f$.

Proof. It suffices to add one point $y\in(x_{i-1},x_i)$: the infima of f over $[x_{i-1},y]$ and $[y,x_i]$ are $\ge m_i$, so L can only increase; similarly U decreases. Then $L(f,P)\le L(f,P\cup P')\le U(f,P\cup P')\le U(f,P')$. ■

Proposition 5.2 (5.1.2, Riemann's criterion)

A bounded f is integrable iff for every $\epsilon>0$ there is a partition P with $U(f,P)-L(f,P)<\epsilon$.

Proof. If the condition holds, $0\le\overline{\int} f-\underline{\int} f\le U(f,P)-L(f,P)<\epsilon$. Conversely, if $f\in\mathcal{R}$, choose P_1,P_2 with $U(f,P_1)<\int f+\epsilon/2$ and $L(f,P_2)>\int f-\epsilon/2$; then $P=P_1\cup P_2$ works by Proposition 5.1. ■

Theorem 5.3 (5.1.3)

If $f,g\in\mathcal{R}$ and $c\in\mathbb{R}$, then $f+g, cf\in\mathcal{R}$ with $\int(f+g)=\int f+\int g$ and $\int cf=c\int f$; and $f\le g$ implies $\int f\le\int g$.

Proof. On each subinterval $\sup(f+g)\le\sup f+\sup g$ and $\inf(f+g)\ge\inf f+\inf g$, so

$$L(f,P)+L(g,P)\le L(f+g,P)\le U(f+g,P)\le U(f,P)+U(g,P).$$

Choosing P with $U - L < \epsilon/2$ for both f and g (common refinement) gives integrability and the formula. For $c \geq 0$, $U(cf, P) = cU(f, P)$ and $L(cf, P) = cL(f, P)$; for $c = -1$, $U(-f, P) = -L(f, P)$. If $f \leq g$, then $U(f, P) \leq U(g, P)$ for all P. ■

Products of integrable functions are integrable: $f^2 \in \mathcal{R}$ because on each subinterval $\sup f^2 - \inf f^2 \leq 2K(M_i - m_i)$ with $K = \sup|f|$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

Theorem 5.4 (5.1.4)

If $f \in \mathcal{R}[a, b]$ and $a < c < b$, then $f \in \mathcal{R}[a, c]$, $f \in \mathcal{R}[c, b]$ and $\int_a^b f = \int_a^c f + \int_c^b f$. Conversely, integrability on $[a, c]$ and $[c, b]$ implies integrability on $[a, b]$.

Proof. For partitions P of $[a, b]$ containing c, with $P_1 = P \cap [a, c]$ and $P_2 = P \cap [c, b]$, we have $U(f, P) = U(f, P_1) + U(f, P_2)$, likewise for L. If $U(f, P) - L(f, P) < \epsilon$ (we may add c to P), the same holds for P_1 and P_2. Conversely, combine partitions of the two halves. Taking infima of upper sums gives the formula. ■

With the convention $\int_b^a f = -\int_a^b f$, the formula holds for any order of a, b, c.

Theorem 5.5 (5.1.5)

If $f \in \mathcal{R}$, then $|f| \in \mathcal{R}$ and $|\int_a^b f| \leq \int_a^b |f|$.

Proof. From $||f(x)| - |f(y)|| \leq |f(x) - f(y)|$, the oscillation of $|f|$ on each subinterval is at most that of f, so $U(|f|, P) - L(|f|, P) \leq U(f, P) - L(f, P)$. Then $-|f| \leq f \leq |f|$ and Theorem 5.3 give the inequality. ■

Example 5.1

The Dirichlet function (1 on $\mathbb{Q}$, 0 elsewhere) is not integrable on $[0, 1]$: every $U(f, P) = 1$ and every $L(f, P) = 0$. Integrability of $|f|$ does not imply that of f: take $f = 1$ on $\mathbb{Q}$ and -1 elsewhere.

5.2 Integrable functions

Motivation. Which functions are integrable? Continuous and monotone ones are, and integrability survives sums, products and composition with continuous functions. These results let us integrate everything we meet in practice without checking upper and lower sums each time.

Theorem 5.6 (5.2.1)

Every continuous $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. f is uniformly continuous (Theorem 3.18): choose δ with $|f(x) - f(y)| < \epsilon/(b-a)$ when $|x-y| < \delta$. If $\|P\| < \delta$, then $M_i - m_i < \epsilon/(b-a)$ (the extrema are attained), so $U(f, P) - L(f, P) < \epsilon$. ■

Theorem 5.7 (5.2.2)

Every monotone $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. Let f increase and P divide $[a,b]$ into n equal parts. Then $M_i = f(x_i)$, $m_i = f(x_{i-1})$, and $U(f,P) - L(f,P) = \frac{b-a}{n}\sum(f(x_i) - f(x_{i-1})) = \frac{b-a}{n}(f(b) - f(a)) < \epsilon$ for large n. ∎

More generally, a bounded function with finitely many discontinuities is integrable (enclose them in short intervals), and so is one whose discontinuity set has measure zero (Chapter 10 of the second volume).

A **Riemann sum** is $S(f,P) = \sum f(t_i)\Delta x_i$ with $t_i \in [x_{i-1}, x_i]$.

Theorem 5.8 (5.2.3)

For bounded f and $A \in \mathbb{R}$ the following are equivalent:
(a) $f \in \mathcal{R}$ and $\int_a^b f = A$;
(b) for every $\epsilon > 0$ there is $\delta > 0$ with $|S(f,P) - A| < \epsilon$ for every partition with $\|P\| < \delta$ and every choice of the t_i.

Proof. (a)⇒(b): let $K = \sup|f|$ and choose P_0, with k points, such that $U(f,P_0) - L(f,P_0) < \epsilon/2$. Let $\|P\| < \delta = \epsilon/(8Kk)$. Passing from P to $P \cup P_0$ changes only the at most k subintervals of P that contain points of P_0 in their interior, and each change of U or L is at most $2K\|P\|$. So $U(f,P) - L(f,P) \le U(f,P\cup P_0) - L(f,P\cup P_0) + 4Kk\delta < \epsilon$. Since both $S(f,P)$ and A lie in $[L(f,P), U(f,P)]$, $|S(f,P) - A| < \epsilon$.

(b)⇒(a): fix P with $\|P\| < \delta$. Choosing the t_i with $f(t_i) > M_i - \epsilon/(b-a)$ gives $U(f,P) < S(f,P) + \epsilon < A + 2\epsilon$; similarly $L(f,P) > A - 2\epsilon$. So $U(f,P) - L(f,P) < 4\epsilon$, and $\int f = A$. ∎

Example 5.2

$\int_0^1 x^2\,dx = \lim_n \frac{1}{n}\sum_{k=1}^n (k/n)^2 = \lim \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}$. Also $\lim_n \sum_{k=1}^n \frac{1}{n+k} = \int_0^1 \frac{dx}{1+x} = \log 2$.

5.3 The fundamental theorem of calculus

Motivation. Differentiation and integration are inverse operations. This connects the area problem with antiderivatives and yields integration by parts and change of variables.

For $f \in \mathcal{R}[a,b]$ let $F(x) = \int_a^x f$.

Theorem 5.9 (5.3.1)

F is uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 and $F'(x_0) = f(x_0)$.

Proof. With $K = \sup|f|$, $|F(y) - F(x)| = |\int_x^y f| \le K|y - x|$. If f is continuous at x_0, given ϵ choose δ with $|f(t) - f(x_0)| < \epsilon$ for $|t - x_0| < \delta$; then for $0 < |h| < \delta$,

$$\left|\frac{F(x_0+h) - F(x_0)}{h} - f(x_0)\right| = \left|\frac{1}{h}\int_{x_0}^{x_0+h} (f(t) - f(x_0))\,dt\right| \le \epsilon.$$

∎

Corollary 5.10 (5.3.2)

Every continuous $f : [a,b] \to \mathbb{R}$ has a primitive, namely $F(x) = \int_a^x f$.

Theorem 5.11 (5.3.3, fundamental theorem of calculus)

If $F:[a,b]\to\mathbb{R}$ is differentiable and $f=F'$ is integrable, then $\int_a^b f = F(b)-F(a)$.

Proof. For any partition, the mean value theorem gives $t_i\in(x_{i-1},x_i)$ with $F(x_i)-F(x_{i-1}) = f(t_i)\Delta x_i$, so $F(b)-F(a)=\sum f(t_i)\Delta x_i \in [L(f,P),U(f,P)]$. As $f\in\mathcal{R}$, $F(b)-F(a)=\int f$. ∎

The integrability hypothesis cannot be dropped: there are differentiable functions with bounded, non-integrable derivatives (Volterra), and $x^2\sin(1/x^2)$ has an unbounded derivative on $[0,1]$.

Proposition 5.12 (5.3.4, change of variables)

If $g:[a,b]\to[c,d]$ is a surjective C^1 function and $f:[c,d]\to\mathbb{R}$ is continuous, then $\int_{g(a)}^{g(b)} f(x)\,dx = \int_a^b f(g(t))g'(t)\,dt$.

Proof. Let F be a primitive of f (Corollary 5.10). By the chain rule $(F\circ g)' = (f\circ g)g'$, which is continuous, hence integrable, so by Theorem 5.11 the right side is $F(g(b))-F(g(a))$, which is also the left side. ∎

Proposition 5.13 (5.3.5, integration by parts)

If $F,G:[a,b]\to\mathbb{R}$ are differentiable and $f=F'$, $g=G'$ are integrable, then $\int_a^b Fg = F(b)G(b)-F(a)G(a)-\int_a^b fG$.

Proof. F,G are continuous, hence integrable, so $(FG)'=fG+Fg$ is integrable (products of integrable functions); apply Theorem 5.11 to FG. ∎

5.4 Functions of bounded variation

Motivation. Which functions are the difference of two increasing functions? Exactly those whose total up-and-down movement is finite. Such functions are the natural integrators in the Riemann–Stieltjes integral and describe rectifiable curves.

For $f:[a,b]\to\mathbb{R}$ and a partition P, let $V(f,P)=\sum|f(x_i)-f(x_{i-1})|$. The **total variation** is $V_a^b(f)=\sup_P V(f,P)$; f is of **bounded variation** if it is finite. Refining P increases $V(f,P)$ (triangle inequality). Monotone functions have $V_a^b(f)=|f(b)-f(a)|$; Lipschitz functions with constant K have $V_a^b(f)\le K(b-a)$; BV functions are bounded, $|f(x)|\le|f(a)|+V_a^b(f)$.

Proposition 5.14 (5.4.1)

If f,g are of bounded variation, then so are $f+g$ and fg, with

$$V(f+g)\le V(f)+V(g),\qquad V(fg)\le\|f\|_\infty V(g)+\|g\|_\infty V(f).$$

If moreover $|f|\ge c>0$, then $1/f$ is of bounded variation with $V(1/f)\le V(f)/c^2$.

Proof. Termwise, $|(f+g)(x_i)-(f+g)(x_{i-1})|\le|\Delta f_i|+|\Delta g_i|$, and

$$|f(x_i)g(x_i)-f(x_{i-1})g(x_{i-1})|\le|f(x_i)||\Delta g_i|+|g(x_{i-1})||\Delta f_i|,$$

and $|1/f(x_i)-1/f(x_{i-1})| = |\Delta f_i|/|f(x_i)f(x_{i-1})|\le|\Delta f_i|/c^2$. Sum and take suprema. ∎

Theorem 5.15 (5.4.2)

If f is of bounded variation on $[a,b]$ and $a<c<b$, then f is of bounded variation on $[a,c]$ and $[c,b]$, and $V_a^b(f)=V_a^c(f)+V_c^b(f)$. Conversely, bounded variation on both halves implies it on $[a,b]$.

Proof. For partitions $P\ni c$ of $[a,b]$, $V(f,P)=V(f,P\cap[a,c])+V(f,P\cap[c,b])$, and every partition may be refined to contain c. Taking suprema gives the formula, both sides being finite or infinite together. ∎

Theorem 5.16 (5.4.3, Jordan decomposition)

$f:[a,b]\to\mathbb{R}$ is of bounded variation iff $f=g-h$ with g,h increasing.

Proof. Monotone functions are of bounded variation, hence so are their differences (Proposition 5.14). Conversely, let $g(x)=V_a^x(f)$ (with $g(a)=0$) and $h=g-f$. g increases by Theorem 5.15, and for $x<y$, $h(y)-h(x)=V_x^y(f)-(f(y)-f(x))\geq 0$. ∎

Consequently a BV function has one-sided limits everywhere and at most countably many discontinuities (Theorem 3.21, Corollary 3.22). A continuous function need not be of bounded variation: $x\sin(1/x)$ on $[0,1]$.

A **curve** is a continuous $\alpha:[a,b]\to\mathbb{R}^m$; its **length** is $\Lambda(\alpha)=\sup_P\sum\|\alpha(x_i)-\alpha(x_{i-1})\|$.

Proposition 5.17 (5.4.4)

For a curve $\alpha=(\alpha_1,\dots,\alpha_m)$ the following are equivalent:
(a) $\Lambda(\alpha)<\infty$;
(b) each α_k is of bounded variation.

Proof. For each k, $|\alpha_k(x_i)-\alpha_k(x_{i-1})|\leq\|\alpha(x_i)-\alpha(x_{i-1})\|\leq\sum_j|\alpha_j(x_i)-\alpha_j(x_{i-1})|$. Summing, $V(\alpha_k,P)\leq\sum_i\|\Delta\alpha_i\|\leq\sum_j V(\alpha_j,P)$; take suprema. ∎

Theorem 5.18 (5.4.5)

Every C^1 curve $\alpha:[a,b]\to\mathbb{R}^m$ has finite length $\Lambda(\alpha)=\int_a^b\|\alpha'(t)\|\,dt$.

Proof. For vector-valued v (integrated componentwise), $\|\int v\|\leq\int\|v\|$: with $u=\int v$, $\|u\|^2=\int\langle u,v\rangle\leq\|u\|\int\|v\|$ by Cauchy–Schwarz. Hence $\|\alpha(x_i)-\alpha(x_{i-1})\|=\|\int_{x_{i-1}}^{x_i}\alpha'\|\leq\int_{x_{i-1}}^{x_i}\|\alpha'\|$, and $\Lambda(\alpha)\leq\int\|\alpha'\|$. Conversely, α' is uniformly continuous; for $\|P\|$ small, $\|\alpha'(t)-\alpha'(x_i)\|<\epsilon$ on $[x_{i-1},x_i]$, so

$$\int_{x_{i-1}}^{x_i}\|\alpha'\|\leq\|\alpha'(x_i)\|\Delta x_i+\epsilon\Delta x_i=\Big\|\int_{x_{i-1}}^{x_i}(\alpha'(t)+\alpha'(x_i)-\alpha'(t))dt\Big\|+\epsilon\Delta x_i\leq\|\Delta\alpha_i\|+2\epsilon\Delta x_i.$$

Summing, $\int\|\alpha'\|\leq\Lambda(\alpha)+2\epsilon(b-a)$. ∎

5.5 The Riemann–Stieltjes integral

Motivation. Replacing lengths Δx by increments $\Delta\alpha$ of a monotone function lets one integral handle sums, ordinary integrals, and expectations of random variables with any distribution.

Let $\alpha:[a,b]\to\mathbb{R}$ be increasing. With $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})$ define $U(f,P,\alpha)=\sum M_i\Delta\alpha_i$, $L(f,P,\alpha)=\sum m_i\Delta\alpha_i$, and $S(f,P,\alpha)=\sum f(t_i)\Delta\alpha_i$. As in Proposition 5.1, refinement decreases U and increases L. $f\in\mathcal{R}(\alpha)$ if $\inf_P U(f,P,\alpha)=\sup_P L(f,P,\alpha)$; the common value is $\int_a^b f\,d\alpha$.

Theorem 5.19 (5.5.1)

Let α,β be increasing and f,g bounded.
(a) If $f,g\in\mathcal{R}(\alpha)$ and $c\in\mathbb{R}$, then $f+g, cf\in\mathcal{R}(\alpha)$ with $\int(f+g)\,d\alpha=\int f\,d\alpha+\int g\,d\alpha$ and $\int cf\,d\alpha=c\int f\,d\alpha$; and $f\le g$ implies $\int f\,d\alpha\le\int g\,d\alpha$.
(b) If $f\in\mathcal{R}(\alpha)$ on $[a,b]$ and $a<c<b$, then $f\in\mathcal{R}(\alpha)$ on $[a,c]$ and $[c,b]$, and $\int_a^b=\int_a^c+\int_c^b$.
(c) If $f\in\mathcal{R}(\alpha)\cap\mathcal{R}(\beta)$ and $c\ge0$, then $f\in\mathcal{R}(\alpha+\beta)\cap\mathcal{R}(c\alpha)$ with $\int f\,d(\alpha+\beta)=\int f\,d\alpha+\int f\,d\beta$ and $\int f\,d(c\alpha)=c\int f\,d\alpha$.
(d) If $f\in\mathcal{R}(\alpha)$, then $|f|\in\mathcal{R}(\alpha)$ and $\left|\int f\,d\alpha\right|\le\int|f|\,d\alpha\le\sup|f|\,(\alpha(b)-\alpha(a))$.

Proof. The proofs of Theorems 5.3, 5.4 and 5.5 apply verbatim with Δx_i replaced by $\Delta\alpha_i\ge0$. For (c), $U(f,P,\alpha+\beta)=U(f,P,\alpha)+U(f,P,\beta)$ and likewise for L. ■

Theorem 5.20 (5.5.2)

For bounded f and increasing α the following are equivalent:
(a) $f\in\mathcal{R}(\alpha)$;
(b) for every $\epsilon>0$ there is P with $U(f,P,\alpha)-L(f,P,\alpha)<\epsilon$;
(c) there is A such that for every $\epsilon>0$ some P_0 has $|S(f,P,\alpha)-A|<\epsilon$ for every $P\supset P_0$ and every choice of tags. In that case $A=\int f\,d\alpha$.

Proof. (a)⇔(b) as in Proposition 5.2.

(b)⇒(c): with $A=\int f\,d\alpha$ and P_0 as in (b), for $P\supset P_0$ both $S(f,P,\alpha)$ and A lie in $[L(f,P,\alpha),U(f,P,\alpha)]$, an interval of length $<\epsilon$.

(c)⇒(b): for $P\supset P_0$, taking suprema and infima over tags, $U(f,P,\alpha)\le A+\epsilon$ and $L(f,P,\alpha)\ge A-\epsilon$. ■

Example 5.3

(a) If α is the unit step at $c\in(a,b)$ ($\alpha=0$ on $[a,c]$, 1 on $(c,b]$) and f is continuous at c, then $\int f\,d\alpha=f(c)$. If f and α are both discontinuous from the right at c, f may fail to be integrable.
(b) For a step function α with jumps c_k at points s_k and continuous f, $\int f\,d\alpha=\sum c_k f(s_k)$: sums are a special case of Stieltjes integrals.

For α of bounded variation, write $\alpha=\alpha_1-\alpha_2$ with α_i increasing (Theorem 5.16). If $f\in\mathcal{R}(\alpha_1)\cap\mathcal{R}(\alpha_2)$ for some such decomposition, we say $f\in\mathcal{R}(\alpha)$ and set $\int f\,d\alpha=\int f\,d\alpha_1-\int f\,d\alpha_2$; by Theorem 5.22 below this does not depend on the decomposition. The Riemann–Stieltjes sum $S(f,P,\alpha)$ is defined as before.

Proposition 5.21 (5.5.3)

If f is continuous on $[a,b]$, then $f\in\mathcal{R}(\alpha)$ for every α of bounded variation.

Proof. It suffices to treat increasing α. Given ϵ, by uniform continuity choose P with $M_i-m_i<\epsilon$ for all i; then $U(f,P,\alpha)-L(f,P,\alpha)\le\epsilon(\alpha(b)-\alpha(a))$. ■

Theorem 5.22 (5.5.4)

For bounded f and α of bounded variation the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = A$;
(b) for every $\epsilon > 0$ there is P_0 with $|S(f,P,\alpha) - A| < \epsilon$ for all $P \supset P_0$ and all tags.

Proof. (a)⇒(b): $S(f,P,\alpha) = S(f,P,\alpha_1) - S(f,P,\alpha_2)$; apply Theorem 5.20(c) to each and take a common refinement. In particular A is determined by f and α, so the integral is well defined.

(b)⇒(a): let $V(x) = V_a^x(\alpha)$; we show $f \in \mathcal{R}(V)$. Let $K = \sup|f|$ and $\epsilon > 0$. Choose P_0 as in (b) and refine it so that $V(b) - \sum_i |\Delta\alpha_i| < \epsilon$ (definition of $V(b)$; refining keeps this). For $P \supset P_0$ write

$$U(f,P,V) - L(f,P,V) = \sum (M_i - m_i)(\Delta V_i - |\Delta\alpha_i|) + \sum (M_i - m_i)|\Delta\alpha_i|.$$

Since $\Delta V_i \geq |\Delta\alpha_i|$, the first sum is at most $2K\epsilon$. For the second, choose s_i, t_i in the i-th interval with $f(t_i) - f(s_i) > M_i - m_i - \epsilon$, and swap them where $\Delta\alpha_i < 0$; then the second sum is less than $\sum (f(t_i) - f(s_i))\Delta\alpha_i + \epsilon V(b) = S(f,P,\alpha) - S'(f,P,\alpha) + \epsilon V(b) < 2\epsilon + \epsilon V(b)$, where S, S' are sums with tags t_i, s_i. By Theorem 5.20, $f \in \mathcal{R}(V)$. Then the sums $S(f,P,V-\alpha) = S(f,P,V) - S(f,P,\alpha)$ satisfy the condition of Theorem 5.20(c) for the increasing function $V - \alpha$, so $f \in \mathcal{R}(V-\alpha)$, and with the decomposition $\alpha = V - (V - \alpha)$, $\int f\,d\alpha = A$. ∎

Theorem 5.23 (5.5.5)

Let α be differentiable with α' continuous on $[a,b]$. If f is bounded and $f \in \mathcal{R}(\alpha)$, then $f\alpha'$ is Riemann integrable and $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. α is of bounded variation (Lipschitz). By the mean value theorem $\Delta\alpha_i = \alpha'(u_i)\Delta x_i$ with $u_i \in (x_{i-1}, x_i)$, so

$$\left| S(f,P,\alpha) - \sum f(t_i)\alpha'(t_i)\Delta x_i \right| \leq K \sum |\alpha'(u_i) - \alpha'(t_i)|\Delta x_i.$$

By uniform continuity of α' this is $\leq K\epsilon(b-a)$ once $\|P\|$ is small. Combining with Theorem 5.22(b), the Riemann sums of $f\alpha'$ over all sufficiently fine refinements of some P_0 are within $\epsilon + K\epsilon(b-a)$ of $\int f\,d\alpha$. By Theorem 5.20 with $\alpha(x) = x$, $f\alpha' \in \mathcal{R}$ with the stated integral. ∎

Theorem 5.24 (5.5.6, integration by parts)

Let f, α be of bounded variation on $[a,b]$. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ and $\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a)$.

Proof. Let $A = \int f\,d\alpha$ and P_0 as in Theorem 5.22(b). Let $Q \supset P_0$ be a partition $a = y_0 < \cdots < y_m = b$ with tags $s_k \in [y_{k-1}, y_k]$. By summation by parts,

$$S(\alpha,Q,f) = \sum_k \alpha(s_k)(f(y_k) - f(y_{k-1})) = f(b)\alpha(b) - f(a)\alpha(a) - \sum_{k=0}^{m} f(y_k)(\alpha(s_{k+1}) - \alpha(s_k)),$$

with $s_0 = a$, $s_{m+1} = b$. The last sum is a Riemann–Stieltjes sum $S(f,P,\alpha)$ for the partition P formed by the points s_k together with the y_k (zero-length intervals contribute nothing), with tags y_k; $P \supset Q \supset P_0$. So $|S(\alpha,Q,f) - (f(b)\alpha(b) - f(a)\alpha(a) - A)| < \epsilon$, and Theorem 5.22 applies to α and f. ∎

Corollary 5.25 (5.5.7, second mean value theorem)

If $f : [a, b] \to \mathbb{R}$ is increasing and g is continuous, there is $c \in [a, b]$ with $\int_a^b fg = f(a) \int_a^c g + f(b) \int_c^b g$.

Proof. Let $G(x) = \int_a^x g$, a C^1 function with $G' = g$. $G \in \mathcal{R}(f)$ (Proposition 5.21), so $f \in \mathcal{R}(G)$ and $\int f\, dG = f(b)G(b) - \int G\, df$ (Theorem 5.24), while $\int f\, dG = \int fg$ (Theorem 5.23). With m, M the minimum and maximum of G, $m(f(b) - f(a)) \le \int G\, df \le M(f(b) - f(a))$, so $\int G\, df = G(c)(f(b) - f(a))$ for some c by the intermediate value theorem. Hence $\int fg = f(b)G(b) - G(c)(f(b) - f(a)) = f(a)G(c) + f(b)(G(b) - G(c))$. ■